

Roots (Part I)

*For all problems in this class, assume that unknowns are positive. Therefore, you do not have to use absolute value symbols when finding roots (e.g. $\sqrt{x^2} = |x|$).

Root Rules

$$\sqrt{ab} = \sqrt{a} \sqrt{b}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Perfect Squares

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

Example I) $\sqrt{24} = \sqrt{4 \cdot 6} = \sqrt{4} \cdot \sqrt{6} = 2 \cdot \sqrt{6} = \boxed{2\sqrt{6}}$

Example II) $3\sqrt{56} = 3\sqrt{4 \cdot 14} = 3\sqrt{4} \cdot \sqrt{14} = 3 \cdot 2 \cdot \sqrt{14} = \boxed{6\sqrt{14}}$

Example III) $\sqrt{27} = \sqrt{9 \cdot 3} = \sqrt{9} \cdot \sqrt{3} = 3 \cdot \sqrt{3} = \boxed{3\sqrt{3}}$

Example IV) $9\sqrt{75} = 9\sqrt{25 \cdot 3} = 9 \cdot \sqrt{25} \cdot \sqrt{3} = 9 \cdot 5 \cdot \sqrt{3} = \boxed{45\sqrt{3}}$

Simplify.

$$\begin{aligned} \textcircled{1} \sqrt{18x^3y^2} &= \sqrt{9 \cdot 2 \cdot x^2 \cdot x \cdot y^2} = \sqrt{9} \cdot \sqrt{2} \cdot \sqrt{x^2} \cdot \sqrt{x} \cdot \sqrt{y^2} \\ &= 3 \cdot \sqrt{2} \cdot x \cdot \sqrt{x} \cdot y \\ &= \boxed{3xy\sqrt{2x}} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \sqrt{32x^7y^3} &= \sqrt{16 \cdot 2 \cdot x^6 \cdot x \cdot y^2 \cdot y} \\ &= \sqrt{16} \cdot \sqrt{2} \cdot \sqrt{x^6} \cdot \sqrt{x} \cdot \sqrt{y^2} \cdot \sqrt{y} \\ &= 4 \cdot \sqrt{2} \cdot x^3 \cdot \sqrt{x} \cdot y \cdot \sqrt{y} = \boxed{4x^3y\sqrt{2xy}} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \sqrt{8x^4y^2} &= \sqrt{4 \cdot 2 \cdot x^4y^2} = \sqrt{4} \cdot \sqrt{2} \cdot \sqrt{x^4} \cdot \sqrt{y^2} \\ &= 2 \cdot \sqrt{2} \cdot x^2 \cdot y \\ &= \boxed{2x^2y\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \sqrt{16x^{10}y^{11}} &= \sqrt{16 \cdot x^{10}y^{10}y} = \sqrt{16} \cdot \sqrt{x^{10}} \cdot \sqrt{y^{10}} \cdot \sqrt{y} \\ &= 4 \cdot x^5 \cdot y^5 \cdot \sqrt{y} \\ &= \boxed{4x^5y^5\sqrt{y}} \end{aligned}$$

$$\begin{aligned} \textcircled{5} \sqrt{20x^5y} &= \sqrt{4 \cdot 5 \cdot x^4 \cdot x \cdot y} = \sqrt{4} \cdot \sqrt{5} \cdot \sqrt{x^4} \cdot \sqrt{x} \cdot \sqrt{y} \\ &= 2 \cdot \sqrt{5} \cdot x^2 \cdot \sqrt{x} \cdot \sqrt{y} \\ &= \boxed{2x^2\sqrt{5xy}} \end{aligned}$$

$$\begin{aligned} \textcircled{6} \sqrt{9x^9y^6} &= \sqrt{9 \cdot x^8 \cdot x \cdot y^6} = \sqrt{9} \cdot \sqrt{x^8} \cdot \sqrt{x} \cdot \sqrt{y^6} \\ &= 3 \cdot x^4 \cdot \sqrt{x} \cdot y^3 \\ &= \boxed{3x^4y^3\sqrt{x}} \end{aligned}$$

$$\begin{aligned}
 \textcircled{7} \sqrt{12x^2y^8} &= \sqrt{4 \cdot 3 \cdot x^2 \cdot y^8} = \sqrt{4} \cdot \sqrt{3} \cdot \sqrt{x^2} \cdot \sqrt{y^8} \\
 &= 2 \cdot \sqrt{3} \cdot x \cdot y^4 \\
 &= \boxed{2xy^4\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{8} \sqrt{25x^6y^{13}} &= \sqrt{25x^6y^{12} \cdot y} = \sqrt{25} \cdot \sqrt{x^6} \cdot \sqrt{y^{12}} \cdot \sqrt{y} \\
 &= 5 \cdot x^3 \cdot y^6 \cdot \sqrt{y} \\
 &= \boxed{5x^3y^6\sqrt{y}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Example V)} \sqrt{28} - 3\sqrt{63} &= \sqrt{4 \cdot 7} - 3\sqrt{9 \cdot 7} \\
 &= \sqrt{4} \cdot \sqrt{7} - 3\sqrt{9} \cdot \sqrt{7} \\
 &= 2 \cdot \sqrt{7} - 3 \cdot 3 \cdot \sqrt{7} \\
 &= 2\sqrt{7} - 9\sqrt{7} \\
 &= \boxed{-7\sqrt{7}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Example VI)} 8\sqrt{44} + \sqrt{99} &= 8\sqrt{4 \cdot 11} + \sqrt{9 \cdot 11} \\
 &= 8 \cdot \sqrt{4} \cdot \sqrt{11} + \sqrt{9} \cdot \sqrt{11} \\
 &= 8 \cdot 2 \cdot \sqrt{11} + 3 \cdot \sqrt{11} \\
 &= 16\sqrt{11} + 3\sqrt{11} \\
 &= \boxed{19\sqrt{11}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{9} 2x\sqrt{5x^3} + 10\sqrt{125x^5} \\
 2x\sqrt{5 \cdot x^2 \cdot x} + 10\sqrt{25 \cdot 5 \cdot x^4 \cdot x} \\
 2x\sqrt{5} \cdot \sqrt{x^2} \cdot \sqrt{x} + 10\sqrt{25} \cdot \sqrt{5} \cdot \sqrt{x^4} \cdot \sqrt{x} \\
 2x\sqrt{5} \cdot x \cdot \sqrt{x} + 10 \cdot 5 \cdot \sqrt{5} \cdot x^2 \cdot \sqrt{x} \\
 2x^2\sqrt{5x} + 50x^2\sqrt{5x} \\
 \boxed{52x^2\sqrt{5x}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{10} -6\sqrt{4y^4} + 10y^2 - 2\sqrt{81y^2} + 17y \\
 -6\sqrt{4} \sqrt{y^4} + 10y^2 - 2\sqrt{81} \cdot \sqrt{y^2} + 17y \\
 -6 \cdot 2 \cdot y^2 + 10y^2 - 2 \cdot 9 \cdot y + 17y \\
 -12y^2 + 10y^2 - 18y + 17y \\
 \boxed{-2y^2 - y}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{11} 5x^3\sqrt{40x^2} - x^2\sqrt{10x^4} \\
 5x^3\sqrt{4 \cdot 10 \cdot x^2} - x^2 \cdot \sqrt{10 \cdot x^4} \\
 5x^3\sqrt{4} \cdot \sqrt{10} \cdot \sqrt{x^2} - x^2 \cdot \sqrt{10} \cdot \sqrt{x^4} \\
 5x^3 \cdot 2 \cdot \sqrt{10} \cdot x - x^2 \cdot \sqrt{10} \cdot x^2 \\
 10x^4\sqrt{10} - x^4\sqrt{10} = \boxed{9x^4\sqrt{10}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{12} \quad & 7y^4\sqrt{8y^7} - 3y\sqrt{y^{10}} - 2y^3\sqrt{36y^6} + 6y^7\sqrt{2y} \\
 & 7y^4\sqrt{4 \cdot 2 \cdot y^6 \cdot y} - 3y\sqrt{y^{10}} - 2y^3\sqrt{36 \cdot y^6} + 6y^7\sqrt{2y} \\
 & 7y^4 \cdot \sqrt{4} \cdot \sqrt{2} \cdot \sqrt{y^6} \sqrt{y} - 3y \cdot y^5 - 2y^3 \sqrt{36} \cdot \sqrt{y^6} + 6y^7 \sqrt{2y} \\
 & 7y^4 \cdot 2 \cdot \sqrt{2} \cdot y^3 \cdot \sqrt{y} - 3y^6 - 2y^3 \cdot 6 \cdot y^3 + 6y^7 \sqrt{2y} \\
 & 14y^7\sqrt{2y} - 3y^6 - 12y^6 + 6y^7\sqrt{2y} \\
 & \boxed{20y^7\sqrt{2y} - 15y^6}
 \end{aligned}$$

$$\begin{aligned}
 \text{Example VII) } \sqrt{3} \cdot \sqrt{6} &= \sqrt{3 \cdot 6} = \sqrt{18} = \sqrt{9 \cdot 2} \\
 &= \sqrt{9} \cdot \sqrt{2} \\
 &= \boxed{3\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Example VIII) } \sqrt{15} \cdot \sqrt{3} &= \sqrt{15 \cdot 3} = \sqrt{45} = \sqrt{9 \cdot 5} \\
 &= \sqrt{9} \cdot \sqrt{5} \\
 &= \boxed{3\sqrt{5}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{13} \quad \sqrt{6x^3} \cdot \sqrt{2x^2} &= \sqrt{6x^3 \cdot 2x^2} = \sqrt{12x^5} \\
 &= \sqrt{4 \cdot 3 \cdot x^4 \cdot x} \\
 &= \sqrt{4} \cdot \sqrt{3} \cdot \sqrt{x^4} \cdot \sqrt{x} \\
 &= 2 \cdot \sqrt{3} \cdot x^2 \cdot \sqrt{x} \\
 &= \boxed{2x^2\sqrt{3x}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{14} \quad \sqrt{5y^6} (3\sqrt{5y^2}) &= 3\sqrt{5y^6 \cdot 5y^2} = 3\sqrt{25y^8} \\
 &= 3\sqrt{25} \cdot \sqrt{y^8} \\
 &= 3 \cdot 5 \cdot y^4 \\
 &= \boxed{15y^4}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{15} \quad \sqrt{10x^3} \cdot \sqrt{2x} &= \sqrt{10x^3 \cdot 2x} = \sqrt{20x^4} \\
 &= \sqrt{4 \cdot 5 \cdot x^4} \\
 &= \sqrt{4} \cdot \sqrt{5} \cdot \sqrt{x^4} \\
 &= 2 \cdot \sqrt{5} \cdot x^2 \\
 &= \boxed{2x^2\sqrt{5}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{16} \quad (\sqrt{7x^5})(\sqrt{7x^2}) &= \sqrt{7x^5 \cdot 7x^2} = \sqrt{49x^7} \\
 &= \sqrt{49 \cdot x^6 \cdot x} \\
 &= \sqrt{49} \cdot \sqrt{x^6} \cdot \sqrt{x} \\
 &= 7 \cdot x^3 \cdot \sqrt{x} \\
 &= \boxed{7x^3\sqrt{x}}
 \end{aligned}$$

Example IX) $\frac{2}{\sqrt{5}}$

$$\frac{2 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{2\sqrt{5}}{\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{5}}{\sqrt{25}}$$

$$\boxed{\frac{2\sqrt{5}}{5}}$$

Example X) $\frac{7}{\sqrt{2}}$

$$\frac{7 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{7\sqrt{2}}{\sqrt{2 \cdot 2}}$$

$$\frac{7\sqrt{2}}{\sqrt{4}}$$

$$\boxed{\frac{7\sqrt{2}}{2}}$$

Example XI) $\frac{1}{\sqrt{72}}$

$$\frac{1}{\sqrt{36 \cdot 2}}$$

$$\frac{1}{\sqrt{36} \cdot \sqrt{2}}$$

$$\frac{1}{6 \cdot \sqrt{2}}$$

$$\frac{1 \cdot \sqrt{2}}{6\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{\sqrt{2}}{6\sqrt{2 \cdot 2}}$$

$$\frac{\sqrt{2}}{6\sqrt{4}}$$

$$\frac{\sqrt{2}}{6 \cdot 2}$$

$$\boxed{\frac{\sqrt{2}}{12}}$$

Example XII) $\frac{6}{\sqrt{50}}$

$$\frac{6}{\sqrt{25 \cdot 2}}$$

$$\frac{6}{\sqrt{25} \cdot \sqrt{2}}$$

$$\frac{6}{5\sqrt{2}}$$

$$\frac{6 \cdot \sqrt{2}}{5\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{6\sqrt{2}}{5\sqrt{2 \cdot 2}}$$

$$\frac{6\sqrt{2}}{5\sqrt{4}}$$

$$\frac{6\sqrt{2}}{5 \cdot 2}$$

$$\frac{6\sqrt{2}}{10} = \boxed{\frac{3\sqrt{2}}{5}}$$

Example XIII) $\sqrt{\frac{5}{16}}$

$$\frac{\sqrt{5}}{\sqrt{16}}$$

$$\boxed{\frac{\sqrt{5}}{4}}$$

Example XIV) $\sqrt{\frac{6}{25}}$

$$\frac{\sqrt{6}}{\sqrt{25}}$$

$$\boxed{\frac{\sqrt{6}}{5}}$$

Example XV) $\sqrt{\frac{9}{10}}$

$$\frac{\sqrt{9}}{\sqrt{10}}$$

$$\frac{3}{\sqrt{10}}$$

$$\frac{3 \cdot \sqrt{10}}{\sqrt{10} \cdot \sqrt{10}}$$

$$\frac{3\sqrt{10}}{\sqrt{10 \cdot 10}}$$

$$\frac{3\sqrt{10}}{\sqrt{100}}$$

$$\boxed{\frac{3\sqrt{10}}{10}}$$

Example XVI) $\sqrt{\frac{49}{11}}$

$$\frac{\sqrt{49}}{\sqrt{11}}$$

$$\frac{7}{\sqrt{11}}$$

$$\frac{7 \cdot \sqrt{11}}{\sqrt{11} \cdot \sqrt{11}}$$

$$\frac{7\sqrt{11}}{\sqrt{11 \cdot 11}}$$

$$\frac{7\sqrt{11}}{\sqrt{121}}$$

$$\boxed{\frac{7\sqrt{11}}{11}}$$

Example XVII) $\sqrt{\frac{7}{11}}$ Example XVIII) $\sqrt{\frac{5}{6}}$

$$\frac{\sqrt{7}}{\sqrt{11}}$$

$$\frac{\sqrt{5}}{\sqrt{6}}$$

$$\frac{\sqrt{7} \cdot \sqrt{11}}{\sqrt{11} \cdot \sqrt{11}}$$

$$\frac{\sqrt{5} \cdot \sqrt{6}}{\sqrt{6} \cdot \sqrt{6}}$$

$$\frac{\sqrt{7 \cdot 11}}{\sqrt{11 \cdot 11}}$$

$$\frac{\sqrt{5 \cdot 6}}{\sqrt{6 \cdot 6}}$$

$$\frac{\sqrt{77}}{\sqrt{121}}$$

$$\frac{\sqrt{30}}{\sqrt{36}}$$

$$\boxed{\frac{\sqrt{77}}{11}}$$

$$\boxed{\frac{\sqrt{30}}{6}}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$$

Example XIX) $\sqrt{\frac{12}{5}}$ Example XX) $\sqrt{\frac{20}{3}}$

$$\frac{\sqrt{12}}{\sqrt{5}}$$

$$\frac{\sqrt{20}}{\sqrt{3}}$$

$$\frac{\sqrt{4 \cdot 3}}{\sqrt{5}}$$

$$\frac{\sqrt{4 \cdot 5}}{\sqrt{3}}$$

$$\frac{\sqrt{4 \cdot 3}}{\sqrt{5}}$$

$$\frac{\sqrt{4 \cdot 5}}{\sqrt{3}}$$

$$\frac{2\sqrt{3}}{\sqrt{5}}$$

$$\frac{2\sqrt{5}}{\sqrt{3}}$$

$$\frac{2\sqrt{3} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{2\sqrt{5} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}}$$

$$\frac{2\sqrt{3 \cdot 5}}{\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{5 \cdot 3}}{\sqrt{3 \cdot 3}}$$

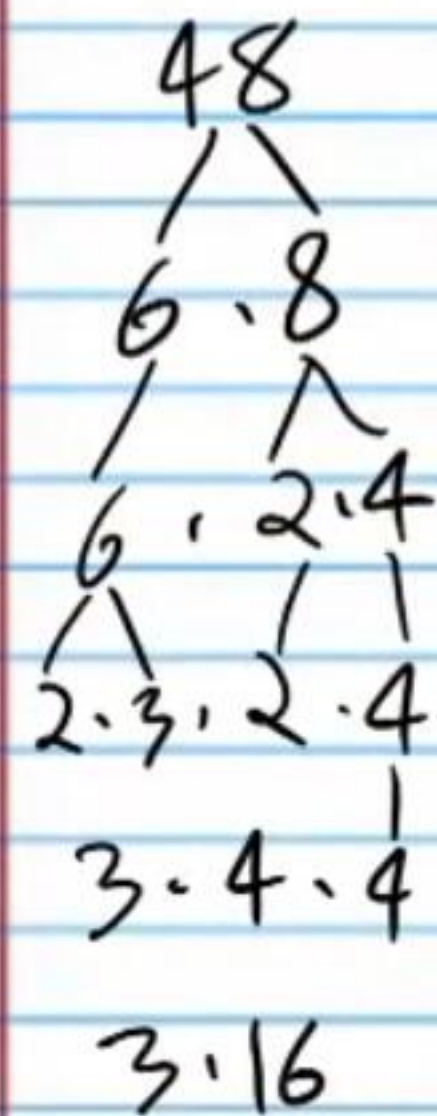
$$\frac{2\sqrt{15}}{\sqrt{25}}$$

$$\frac{2\sqrt{15}}{\sqrt{9}}$$

$$\boxed{\frac{2\sqrt{15}}{5}}$$

$$\boxed{\frac{2\sqrt{15}}{3}}$$

Example XXI) $\sqrt{\frac{5}{48}}$ Example XXII) $\sqrt{\frac{2}{27}}$



$$\frac{\sqrt{5}}{\sqrt{48}}$$

$$\frac{\sqrt{5}}{\sqrt{16 \cdot 3}}$$

$$\frac{\sqrt{5}}{\sqrt{16 \cdot \sqrt{3}}}$$

$$\frac{\sqrt{5}}{4 \cdot \sqrt{3}}$$

$$\frac{\sqrt{5 \cdot \sqrt{3}}}{4 \sqrt{3 \cdot \sqrt{3}}}$$

$$\frac{\sqrt{5 \cdot 3}}{4 \sqrt{3 \cdot 3}}$$

$$\frac{\sqrt{15}}{4 \sqrt{9}}$$

$$\frac{\sqrt{15}}{4 \cdot 3} = \boxed{\frac{\sqrt{15}}{12}}$$

$$\frac{\sqrt{2}}{\sqrt{27}}$$

$$\frac{\sqrt{2}}{\sqrt{27}}$$

$$\frac{\sqrt{2}}{\sqrt{9 \cdot 3}}$$

$$\frac{\sqrt{2}}{\sqrt{9 \cdot \sqrt{3}}}$$

$$\frac{\sqrt{2}}{3 \cdot \sqrt{3}}$$

$$\frac{\sqrt{2 \cdot \sqrt{3}}}{3 \sqrt{3 \cdot \sqrt{3}}}$$

$$\frac{\sqrt{2 \cdot 3}}{3 \sqrt{3 \cdot 3}}$$

$$\frac{\sqrt{6}}{3 \sqrt{9}} = \frac{\sqrt{6}}{3 \cdot 3} = \boxed{\frac{\sqrt{6}}{9}}$$

Example XXIII) $\sqrt{\frac{8}{45}}$ Example XXIV) $\sqrt{\frac{54}{125}}$

$$\frac{\sqrt{8}}{\sqrt{45}}$$

$$\frac{\sqrt{4 \cdot 2}}{\sqrt{9 \cdot 5}}$$

$$\frac{\sqrt{4} \sqrt{2}}{\sqrt{9} \sqrt{5}}$$

$$\frac{2 \sqrt{2}}{3 \sqrt{5}}$$

$$\frac{2 \sqrt{2 \cdot \sqrt{5}}}{3 \sqrt{5 \cdot \sqrt{5}}}$$

$$\frac{2 \sqrt{2 \cdot 5}}{3 \sqrt{5 \cdot 5}}$$

$$\frac{2 \sqrt{10}}{3 \sqrt{25}}$$

$$\frac{2 \sqrt{10}}{3 \cdot 5} = \boxed{\frac{2 \sqrt{10}}{15}}$$

$$\frac{\sqrt{54}}{\sqrt{125}}$$

$$\frac{\sqrt{9 \cdot 6}}{\sqrt{25 \cdot 5}}$$

$$\frac{\sqrt{9 \cdot \sqrt{6}}}{\sqrt{25 \cdot \sqrt{5}}}$$

$$\frac{3 \sqrt{6}}{5 \sqrt{5}}$$

$$\frac{3 \sqrt{6 \cdot \sqrt{5}}}{5 \sqrt{5 \cdot \sqrt{5}}}$$

$$\frac{3 \sqrt{6 \cdot 5}}{5 \sqrt{5 \cdot 5}}$$

$$\frac{3 \sqrt{30}}{5 \sqrt{25}}$$

$$\frac{3 \sqrt{30}}{5 \cdot 5} = \boxed{\frac{3 \sqrt{30}}{25}}$$

Example XXV) $\frac{\sqrt{8}}{\sqrt{32}} = \sqrt{\frac{8}{32}} = \sqrt{\frac{1}{4}} = \frac{\sqrt{1}}{\sqrt{4}}$
 $= \boxed{\frac{1}{2}}$

Example XXVI) $\frac{\sqrt{15}}{\sqrt{12}} = \sqrt{\frac{15}{12}} = \sqrt{\frac{5}{4}}$
 $= \boxed{\frac{\sqrt{5}}{2}}$